Fraud / Anomaly Monitoring & Investigation Dashboard

Background

You are a **Business Analytics Associate** at an e-commerce company. Finance and Operations teams report a rise in:

- suspicious transactions (promo abuse, card testing, fake COD orders)
- high return/refund losses
- unusually high discount usage and repeat cancellations

Leadership wants an **always-on fraud/anomaly monitoring system** that answers:

1. **What are the biggest fraud/anomaly patterns right now?**
2. **Which customers/sessions/orders should be investigated first (risk ranking)?**
3. **What is the estimated loss and how can we reduce it?**
4. **What experiments / controls should be introduced and how will success be measured?**

You will build an end-to-end solution: clean data layer, anomaly rules + scoring, weekly trends, investigation queue, and a dashboard.

> This is an analytics + monitoring capstone (not “hard fraud ML”). Students can optionally build an ML anomaly model, but rule-based + explainable scoring is mandatory.

Datasets Provided (Folder Contents)

You will receive a folder containing:

1. `users.csv` — user profile & signup info
2. `sessions.csv` — session/channel/device metadata (traffic + attribution context)
3. `orders.csv` — order-level data (amounts, payment, address signals, coupon)
4. `order_items.csv` — items per order (SKU/category/qty/unit price)
5. `payments.csv` — payment attempts + results (success/fail, reason codes)
6. `shipments.csv` — shipment events (shipped/delivered/RTO timestamps)

7. `refunds.csv` — refunds/returns (refund amount, reason, approved/rejected)
8. `coupons.csv` — coupon rules (min cart, max discount, allowed categories, new-user only, expiry)
9. `products.json` — product catalog (category, price, cost, rating)

> Note: The data intentionally includes duplicates, missing values, inconsistent casing, and outliers. Cleaning is part of the project.

💡 **Download the dataset from here: https://drive.google.com/drive/folders/1R4y5SqFLIByL0aNinarPHxURZxvrqXMy?usp=sharing**

Business Rules & Definitions (Given)

You must use the following definitions consistently:

Fraud / Anomaly (Project Definition)

An order is “risky” if it shows behaviors such as:

- unusually high discount or repeated coupon use
- high payment failures before success (card testing behavior)
- unusually high order value or quantity vs normal patterns
- address/pincode/device reuse across multiple accounts
- high return/refund probability signals
- suspicious COD patterns (high RTO rate)

Outcomes (to measure success)

- reduced refund losses (₹)
- reduced RTO losses (proxy)
- reduced high-risk orders shipped (count)
- stable conversion (guardrail)

Your Task (End-to-End)

Part A — Problem Framing (Mandatory)

Create a short framing doc with:

- **Business objective**: reduce fraud loss while minimizing false positives
- **North Star metric** (choose one and justify):
  - monthly fraud loss prevented (₹) / net savings (₹)
  - reduction in high-risk orders shipped
- **KPIs (8–12)** must include:
  - refund rate and refund amount (₹)
  - RTO rate (for COD orders)
  - coupon usage rate and average discount%
  - payment failure rate per session/order
  - suspicious orders rate (based on your score threshold)
  - "investigation precision" proxy (how many flagged orders actually refund/RTO)
  - guardrail: overall conversion rate or gross revenue trend
- Stakeholder questions (5–7), e.g. "which coupon is abused the most?"

Part B — Data Architecture & ETL (Python) (Mandatory)

Build a reproducible ETL pipeline that:

1. loads raw datasets
2. fixes duplicates, casing, missing values (document choices)
3. creates curated tables with clear keys & joins

Output 1: `fact_orders_enriched.csv` (1 row per order)

Must include:

- order_id, user_id, session_id, order_ts
- gross_amount, discount_amount, net_amount, payment_method
- coupon_id, coupon_discount_pct
- item_count, total_qty, top_category
- shipping_pincode, shipping_city_tier (if derivable)

- **derived risk signals (at least 10)** such as:
  - high_discount_flag
  - multi_coupon_user_flag (if repeat usage)
  - payment_fail_count_before_success
  - device_reuse_count (from sessions)
  - pincode_reuse_count
  - order_value_zscore_by_category
  - qty_outlier_flag
  - new_user_flag
  - cod_flag
  - late_night_order_flag

Output 2: `fact_user_risk_weekly.csv` (1 row per user per week)

Must include:

- user_id, week_start
- orders_count, net_revenue, refunds_count, refund_amount
- coupon_orders_count, avg_discount_pct
- cod_orders_count, rto_count
- payment_failures_count
- **risk_score_avg** (derived from your risk scoring)

Output 3: `investigation_queue.csv` (Mandatory)

A ranked table for Ops to action:

- order_id, risk_score, risk_band (High/Medium/Low)
- top 3 risk reasons (text labels)
- recommended action (hold/manual review/call verification/auto-cancel)
- evidence fields (the exact values that triggered the reasons)

Part C — Analytics (Insights + KPIs)

1) Fraud Landscape Diagnosis (Weekly)

- trends of refunds, RTO, coupon usage, payment failures
- identify top spikes/weeks and likely causes

2) Top Fraud/Anomaly Patterns (Mandatory)

Identify at least **5 patterns**, and quantify:

- how many orders
- how much ₹ loss (proxy)
- what segment it concentrates in (channel/device/city tier/new vs returning)

Examples of patterns:

- “coupon X used by many new users from same pincode”
- “many payment failures then small successful payment”
- “COD orders from specific region with high RTO”

3) Segment Deep Dive (Mandatory)

Analyze anomalies by:

- channel × device
- new vs returning users
- city tier / pincode clusters
- payment method (COD vs prepaid)
- top categories

Deliver:

- top 3 segments where **refund loss** is highest
- top 3 segments where **risk score** is highest
- recommended controls for each segment

4) Investigation Table (Mandatory)

Create a table for **top 2 fraud patterns**, containing:

- pattern definition
- volume (# orders) and loss proxy (₹)
- where it concentrates (segment/pincode/device)
- 2–3 hypotheses about cause

- 1 follow-up experiment/data needed to validate each hypothesis
- **evidence used** (metrics/charts/tables)

Part D — Risk Scoring System (Must-have)

Create an explainable **risk score** (0–100) per order.

Minimum requirements:

- define **10–15 risk signals**
- assign weights (or use logistic-style scoring) with justification
- output:
  - `risk_score`
  - `risk_band` thresholds (High/Med/Low)
  - top 3 drivers per order (reason codes)

Optional (bonus):

- build an ML model to predict refund/RTO using the engineered features, and compare against rule score.

Part E — Controls + Experiment Design (Mandatory)

Propose **3–5 interventions**, examples:

- hold high-risk orders for manual review
- OTP verification for COD orders above ₹X
- limit coupon usage by device/pincode
- blocklist for repeated payment failures

For each intervention:

- define control/treatment
- primary metric (fraud loss prevented)
- guardrails (conversion, customer complaints proxy)
- expected time window and success threshold

Part F — Impact Estimation (30-Day Projection)

Estimate:

- current monthly loss (refund + RTO proxy)
- expected loss prevented using your queue (base/best/worst)
- operational cost trade-off (manual reviews count)

Part G — Dashboard (Tableau / PowerBI / Excel) (Mandatory)

Create a dashboard with **minimum 4 views/pages**:

View 1: Executive Summary

- fraud KPIs: refund loss, RTO rate, payment fail rate, coupon abuse rate
- week-over-week trend
- total flagged orders + risk-band split

View 2: Fraud Drivers & Patterns

- top patterns table
- heatmaps by channel/device/payment method
- top coupons involved

View 3: Investigation Queue (Operational)

- filter by risk band, city tier, channel, coupon
- searchable table with order_id + top reasons + recommended action

View 4: Controls & Impact

- projected prevented loss (scenario)
- expected manual review volume
- experiment plan summary

Final Narrative (Decision Memo)

Create either:

- **7–10 slide deck**, OR
- **2–3 page memo**

Must include:

1. objective + KPI definitions
2. data pipeline summary + curated tables
3. top insights & patterns (quantified)
4. scoring system (signals + thresholds)
5. investigation queue + examples
6. controls + experiment plan
7. 30-day impact estimate + risks/limitations

Submission Guidelines (What You Must Submit)

Create a folder named:

`<YourName>_Capstone_FraudMonitoring/`

Include:

1) `README.md`

- objective + north star
- definitions (fraud/anomaly, loss proxy, risk score)
- how to run ETL end-to-end
- curated outputs generated
- dashboard tool used + how to open it

2) `/data/` *(generated only)*

- `fact_orders_enriched.csv`
- `fact_user_risk_weekly.csv`
- `investigation_queue.csv`

3) `/etl/`

- `etl_pipeline.py`

4) `/analysis/`

- `analysis.ipynb` / `analysis.py` (KPIs + patterns + scoring + impact)

5) `/dashboard/`

- `.twbx/.twb` OR `.pbix` OR `.xlsx`
- screenshots/pdf export

6) `/final_story/`

- `final_deck.pdf` OR `final_memo.pdf`

Google Drive Submission (Important)

- Upload your full submission folder to Google Drive.
- Share **one public view link** (anyone with the link can view).
- **Check the link in Incognito/Private mode** to ensure it opens without access request.
- Submit that link.